

Analysis and Visualization of Global COVID-19 Data

Final Report

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1. Table of Contents

1. TABLE OF CONTENTS	2
2. PROJECT OVERVIEW	3
3. DATA SET	5
4. FEATURES AND APPROACHES	7
4.1. Interactive Choropleth World Map of Pandemic Severity	7
4.2. Scatter Plot with Linear Regression Analysis of Vaccination vs Mortality	8
4.3. Correlation Analysis Between Economic Level and Vaccination Coverage	8
4.4. Global COVID-19 Time Trend Analysis and Interactive Visualization	10
4.5. Correlation Analysis Between Population Density and COVID-19 Cases	12
5. KEY FINDINGS	14
5.1. Comparison of the severity of the epidemic among different countries	14
5.2. COVID-19 full vaccination rate shows a weak positive correlation with cumulative deaths per million	17
5.3. Correlation Analysis Between Economic Level and Vaccination Coverage	18
5.4. Global COVID-19 Time Trend Analysis	23
5.5. Correlation Analysis Between Population Density and COVID-19 Cases	25
6. CONCLUSION AND DISCUSSION	27
6.1. Core Conclusions	27
6.2. Key Discussion	27
6.3. Limitations and Implications	28



2. Project Overview

This project performs **comprehensive analysis and interactive visualization of global COVID-19 data** to investigate patterns, disparities, and influencing factors of the pandemic across countries and regions.

The main goal is to explore how **economic conditions, population density, vaccination coverage, geographic distribution, and time dynamics** relate to COVID-19 transmission severity and public health outcomes. We aim to reveal global disparities, verify key correlations, and provide data-supported insights for understanding pandemic spread and vaccine equity.

The project uses a cleaned, country-level COVID-19 dataset from Our World in Data, including epidemiological, vaccination, demographic, and economic indicators. We integrate **cross-sectional statistical analysis** and **time-series trend analysis**, with data preprocessing (aggregation, log transformation, outlier handling) and a combination of static and interactive visualizations to ensure comparability and interpretability.

The project includes five core analytical modules:

- Interactive world maps showing pandemic severity and spatial-temporal spread
- Regression analysis of vaccination coverage and mortality
- Correlation analysis between economic level and vaccine equity
- Time-series trend and dashboard visualization of global pandemic waves
- Log-scale regression analysis of population density and infection rates

From the data, we obtained key findings:

- Economic strength strongly predicts vaccination coverage, leading to a significant global gap in vaccine equity.
- Population density has only a very weak positive correlation with infection rates and low explanatory power.
- At the country level, higher vaccination coverage does not clearly correspond to lower mortality, due to confounding factors such as testing capacity and age structure.



- The pandemic shows clear regional clustering and periodic waves, with Europe and the Americas more severely affected in reported data.
- Non-economic factors, including government policies, public health systems, and regional characteristics, play critical roles in pandemic outcomes.



3. Data Set

Since this project utilizes a static dataset for data analysis and machine learning rather than a Database Management System, a traditional Entity-Relationship (ER) diagram is not strictly applicable. Instead, the data schema is defined by a comprehensive flat-file structure (CSV) sourced from "Our World in Data".

The dataset consists of a single primary entity, **COVID_Global_Record**, which captures daily COVID-19 statistics and demographic information for various countries. To ensure clarity and facilitate data processing, the attributes (columns) are logically grouped into the following categories:

1. Geographic and Temporal Attributes

These fields define the spatial and temporal granularity of the data.

- **continent (String):** The continent of the geographical location.
- **country (String):** The name of the country or region.
- **code (String):** The ISO 3166-1 alpha-3 country code.
- **date (Date):** The date of the recorded observation.

2. Confirmed Cases & Deaths (Epidemiological Data)

These fields record the core metrics of the pandemic's spread and mortality.

- **Cases:** `total_cases`, `new_cases`, `new_cases_smoothed`, `total_cases_per_million`, `new_cases_per_million`, `new_cases_smoothed_per_million`.
- **Deaths:** `total_deaths`, `new_deaths`, `new_deaths_smoothed`, `total_deaths_per_million`, `new_deaths_per_million`, `new_deaths_smoothed_per_million`.
- **Excess Mortality:** `excess_mortality`, `excess_mortality_cumulative`, `excess_mortality_cumulative_absolute`, `excess_mortality_cumulative_per_million`.

3. Hospitalization and ICU Data

Metrics indicating the burden on the healthcare system.

- `hosp_patients`, `hosp_patients_per_million`
- `weekly_hosp_admissions`, `weekly_hosp_admissions_per_million`



- `icu_patients`, `icu_patients_per_million`
- `weekly_icu_admissions`, `weekly_icu_admissions_per_million`

4. Testing and Policy Metrics

Data related to government response and testing capacity.

- **Testing:** `total_tests`, `new_tests`, `total_tests_per_thousand`, `new_tests_per_thousand`, `new_tests_smoothed`, `new_tests_smoothed_per_thousand`, `positive_rate`, `tests_per_case`.
- **Policy & Transmission:** `stringency_index` (government response strictness), `reproduction_rate` (R_0 , estimate of transmission).

5. Vaccination Data

Tracking the progress of immunization campaigns.

- **Absolute Numbers:** `total_vaccinations`, `people_vaccinated`, `people_fully_vaccinated`, `total_boosters`, `new_vaccinations`, `new_vaccinations_smoothed`, `new_people_vaccinated_smoothed`.
- **Relative Metrics (per hundred/million):** `total_vaccinations_per_hundred`, `people_vaccinated_per_hundred`, `people_fully_vaccinated_per_hundred`, `total_boosters_per_hundred`, `new_vaccinations_smoothed_per_million`, `new_people_vaccinated_smoothed_per_hundred`.

6. Demographics and Socioeconomic Indicators

Static variables used for cross-country comparisons and correlation analysis.

- **Population:** `population`, `population_density`.
- **Health & Age:** `median_age`, `life_expectancy`, `diabetes_prevalence`.
- **Economy & Infrastructure:** `gdp_per_capita`, `extreme_poverty`, `handwashing_facilities`, `hospital_beds_per_thousand`, `human_development_index`.



4. Features and Approaches

4.1. Interactive Choropleth World Map of Pandemic Severity

This feature visualizes the severity of COVID-19 across countries by mapping cumulative cases per million and cumulative deaths per million onto a world map using a color gradient. The darker the color, the more severely the country has been affected by the pandemic. The map supports hover interactions — when the user hovers over a country, a displays detailed information including country name, continent, total cases per million, and total deaths per million, enabling intuitive cross-country comparison at a glance.

Approaches 1:

Data Cleaning: Load `compact_clean.csv` and filter out aggregated entries (e.g., "World", "Asia", "European Union") by retaining only rows whose code field is a valid 3-letter ISO country code.

Missing Value Handling: Fill missing values in `total_cases_per_million` and `total_deaths_per_million` with 0.

Outlier Treatment: Use the 95th percentile as the upper bound of the color scale to prevent extreme small countries (e.g., San Marino, Andorra) from distorting the overall color distribution.

Rendering: Use Plotly Express's choropleth function with ISO 3-letter codes as the geographic identifier, applying a Reds sequential color scale.

Time-based Pandemic Evolution Animation. Building on 4.1, this feature introduces a temporal dimension.

The pandemic data is sliced by month, producing an animated map with a time slider. Users can drag the slider or click the play button to observe the full evolution of the pandemic — from the early outbreak in early 2020 to its global spread.

Approaches 2:

Time Bucketing: Convert the date field into monthly periods using `pd.to_period('M')`.

Aggregation Strategy: Since `total_cases_per_million` is a cumulative value, group by (country, period) and take the last day's value within each period (`.last()`) as the representative value.

Unified Color Scale: All time frames share the same `range_color`, ensuring color depths across different months are directly comparable.



Animation Generation: Use Plotly's `animation_frame` parameter to automatically generate an animation with playback controls.

4.2. Scatter Plot with Linear Regression Analysis of Vaccination vs Mortality

This feature analyzes the relationship between COVID-19 full vaccination rates and cumulative mortality rates across countries. The main objective is to visually and statistically explore whether higher vaccination coverage is associated with lower mortality outcomes at the country level, providing insights into the potential protective effect of vaccination during the pandemic through scatter plots, regression analysis, and correlation metrics.

Approaches:

Data Loading: Load the cleaned dataset (`compact_clean.csv`) containing time-series COVID-19 data with vaccination and mortality indicators.

Data Preprocessing & Aggregation: Convert the date column to datetime format. Group data by country and compute maximum `people_fully_vaccinated_per_hundred` (peak vaccination coverage) and maximum `total_deaths_per_million` (final cumulative mortality). Filter to include only countries with at least 5% full vaccination rate.

Visualization: Use Seaborn's `replots` to generate the scatter plot with regression line (red), point transparency, and grid. Annotate a sample of countries for clarity.

Statistical Analysis: Calculate Pearson correlation coefficient and p-value to quantify the linear relationship between vaccination rate and mortality.

Rendering: Matplotlib/Seaborn with clear titles, axis labels, and annotation of statistical results directly on the plot. The figure is saved as a high-resolution PNG.

4.3. Correlation Analysis Between Economic Level and Vaccination Coverage

This feature quantifies the statistical relationship between national economic level (measured by GDP per capita) and COVID-19 vaccination coverage, and verifies the core research question: "Did wealthier countries achieve



higher vaccination coverage?". It reveals the impact of economic factors on global vaccine equity and identifies outlier countries that deviate from the general trend through residual analysis.

Approaches 1:

Data Preprocessing: Extract cross-sectional data by sorting records by date and retaining the latest entry for each country; filter out samples with missing values and those with GDP per capita ≤ 0 ; apply logarithmic transformation to GDP per capita to reduce data skewness and improve the linearity of the relationship.

Statistical Methods:

1. Linear regression analysis to calculate slope, intercept, coefficient of determination (R^2) and statistical significance (p-value)
2. Residual analysis to identify the top 5 positive and negative residual outliers (countries with significantly higher or lower vaccination rates than expected based on their economic level)

Visualization Approach: Custom joint regression scatter plot implemented with Matplotlib gridspec, including:

- Main scatter plot with continent color coding
- Regression line with 95% confidence interval
- Marginal histograms showing the independent distribution of the two variables
- Automatic annotation of outlier country names with white background bounding boxes

Approaches 2:

Data Source: Uses exactly the same cleaned cross-sectional dataset as the main analysis to ensure consistency and comparability of results.

Statistical Methods:

1. Calculate descriptive statistics (median, minimum, maximum, interquartile range) of vaccination coverage grouped by continent
2. Apply the $1.5 \times \text{IQR}$ rule for outlier detection to identify countries within each continent whose vaccination rates significantly deviate from the regional average
3. Quantify the specific magnitude of the "continental divide" through inter-continental median comparisons, providing data support for regional differences observed in the correlation analysis



Visualization Approach: Grouped box plot implemented with Seaborn, serving as supplementary visualization material to the main scatter plot:

- Intuitively presents the central tendency and dispersion of vaccination coverage across continents
- Clearly demonstrates the overall level differences between different continents
- Highlights extreme outlier countries within each region

Approaches 3:

This feature visualizes the global geographical distribution of three core variables (at least one dose coverage, full vaccination coverage, GDP per capita) through interactive maps, revealing the spatial aggregation and geographical correlation of variables, and compensates for the deficiency of statistical charts in expressing spatial dimensions.

Data Preprocessing: Match the country ISO code field with Plotly's built-in geographical boundary data; filter out samples with missing ISO codes.

Visualization Approach: Interactive choropleth maps implemented with Plotly Express:

- Adopt the Jenks Natural Breaks classification method to maximize inter-group differences and minimize intra-group differences
- Use differentiated color schemes for different variables: Viridis for at least one dose coverage, Plasma for full vaccination coverage, and Blues for GDP per capita
- Add hover interaction to display country name, ISO code and precise numerical values
- Standardize map projection and dimensions for direct comparison of spatial patterns

4.4. Global COVID-19 Time Trend Analysis and Interactive Visualization

This feature focuses on analyzing and visualizing the global time-series trends of COVID-19 cases and deaths across different countries. The objective is to help users explore how the pandemic evolved over time and compare the severity and progression of COVID-19 among selected countries.

The feature was implemented using the cleaned dataset generated from the preprocessing stage. Key indicators such as `new_cases_smoothed_per_million` and `new_deaths_smoothed_per_million` were used to reduce daily reporting fluctuations and allow fair comparisons between countries with different population sizes. The analysis



was first developed in a Jupyter Notebook environment for exploratory data analysis (EDA) and later extended into an interactive Streamlit dashboard for dynamic visualization and user interaction.

Several interactive visualizations were developed in this feature, including:

- Daily new confirmed COVID-19 cases trend across multiple countries
- Daily new confirmed COVID-19 deaths trend across multiple countries
- Cases versus deaths comparison for a selected country

The dashboard allows users to:

- Dynamically select countries for comparison
- Switch between different chart types
- Explore trends using interactive hover tooltips
- Use a time-range slider to zoom into specific periods

Approaches:

The implementation mainly uses Pandas for data processing, Plotly for interactive visualization, and Streamlit for dashboard deployment. Pandas was used to load and filter the cleaned COVID-19 dataset, convert the date column into datetime format, and dynamically select countries for comparative analysis. The study primarily used `new_cases_smoothed_per_million` and `new_deaths_smoothed_per_million` to reduce daily reporting fluctuations and enable fair comparisons between countries with different population sizes. Plotly Express line charts were adopted to visualize long-term COVID-19 trends with interactive hover tooltips and time-range sliders, while Streamlit was used to transform the notebook-based exploratory analysis into a user-friendly interactive dashboard that supports country selection and chart switching without modifying the source code.



4.5. Correlation Analysis Between Population Density and COVID-19 Cases

This feature quantifies the statistical relationship between national population density and COVID-19 infection severity (measured by total cases per million), and verifies the core research question: *“Did countries with higher population density have more severe COVID-19 outbreaks?”*. It reveals the influence of population aggregation on pandemic spread and detects outliers that deviate from the general trend.

Approaches:

Data Preprocessing: Extract the latest cross-sectional data for each country by date; filter invalid and missing values; apply **logarithmic transformation** to population density to reduce skewness and improve trend visibility.

Statistical Methods:

1. Linear regression to compute slope, intercept, R^2 , and p-value
2. Pearson correlation to evaluate relationship strength
3. Residual analysis to identify outlier countries

Visualization: Log-scale regression scatter plot using Seaborn & Matplotlib:

- Main scatter plot colored by continent (tab10 palette)
- Logarithmic x-axis for balanced distribution display
- Red regression line with 95% confidence interval
- Automatic annotation of top outliers
- Statistics (N, R^2 , p) displayed in the title

Data Source: Uses the same cleaned global dataset to ensure consistency with other analyses in the project.

Visualization Features:

- Consistent figure style, font, and layout with the group’s theme
- Semi-transparent points to avoid overlap
- Grid lines for improved readability
- High-resolution image export for reports and PPT

This visualization clearly shows the weak positive correlation between population density and infection rates, and supports further analysis of non-density factors such as policy, health systems, and regional differences.

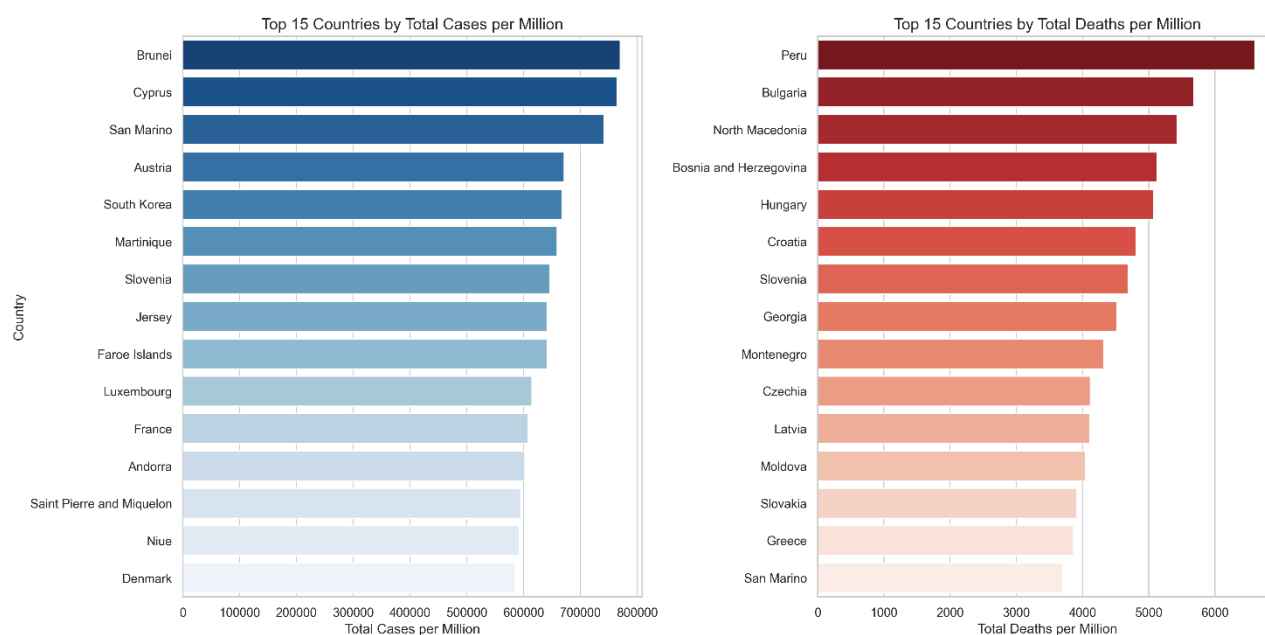




5. Key Findings

5.1. Comparison of the severity of the epidemic among different countries

Pandemic severity correlates with geography and development level. The world map reveals that countries with the highest cases per million are concentrated in Europe (e.g., Austria, France, Denmark) and the Americas (e.g., the United States) — regions characterized by developed economies, high population mobility, and strong testing infrastructure. Conversely, several African countries report low case counts, but this likely reflects insufficient testing coverage rather than genuinely low infection rates.

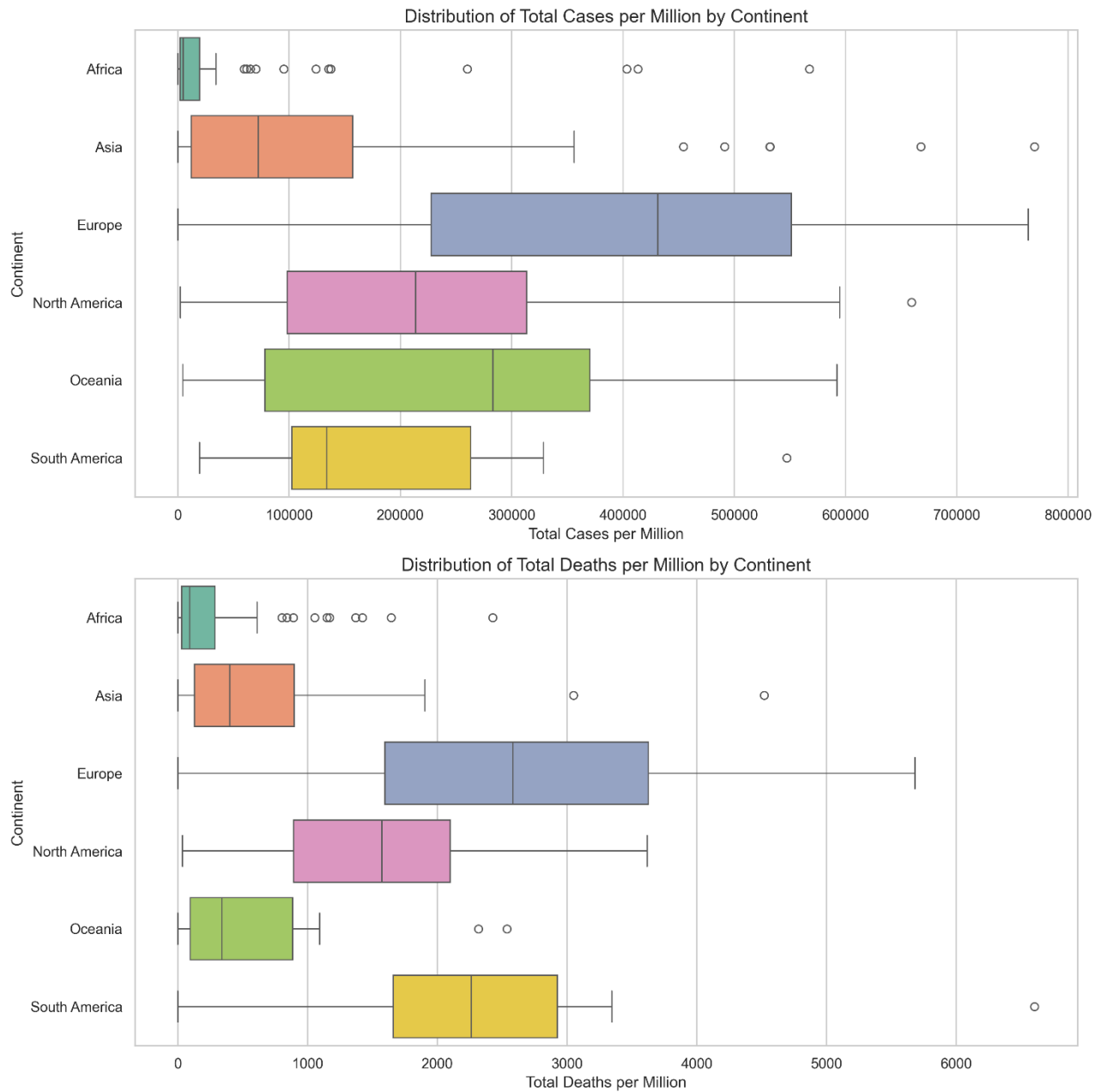


The pandemic spread followed clear temporal phases. The animated map reveals distinct stages of global spread: from January to March 2020, the outbreak was concentrated in China and East Asia; between March and June 2020, it rapidly expanded into Europe and North America, with map colors deepening sharply during this window; in late 2020 through 2021, the wave extended to South America, India, and Eastern Europe; and in early 2022, the Omicron variant triggered a near-simultaneous global surge, with virtually all countries darkening at once.



Continental comparison shows Europe most affected and Africa lowest in reported figures. The line chart shows that Europe's cumulative cases per million remained consistently higher than other continents — likely linked to an aging population and dense cross-border mobility. Africa's curve, in contrast, sits visibly at the bottom, again reinforcing the influence of testing capacity disparities. North America experienced an early and sharp peak in death rates, while South America saw a pronounced death-rate spike in mid-2021 (associated with variants such as Gamma).





5.2. COVID-19 full vaccination rate shows a weak positive correlation with cumulative deaths per million

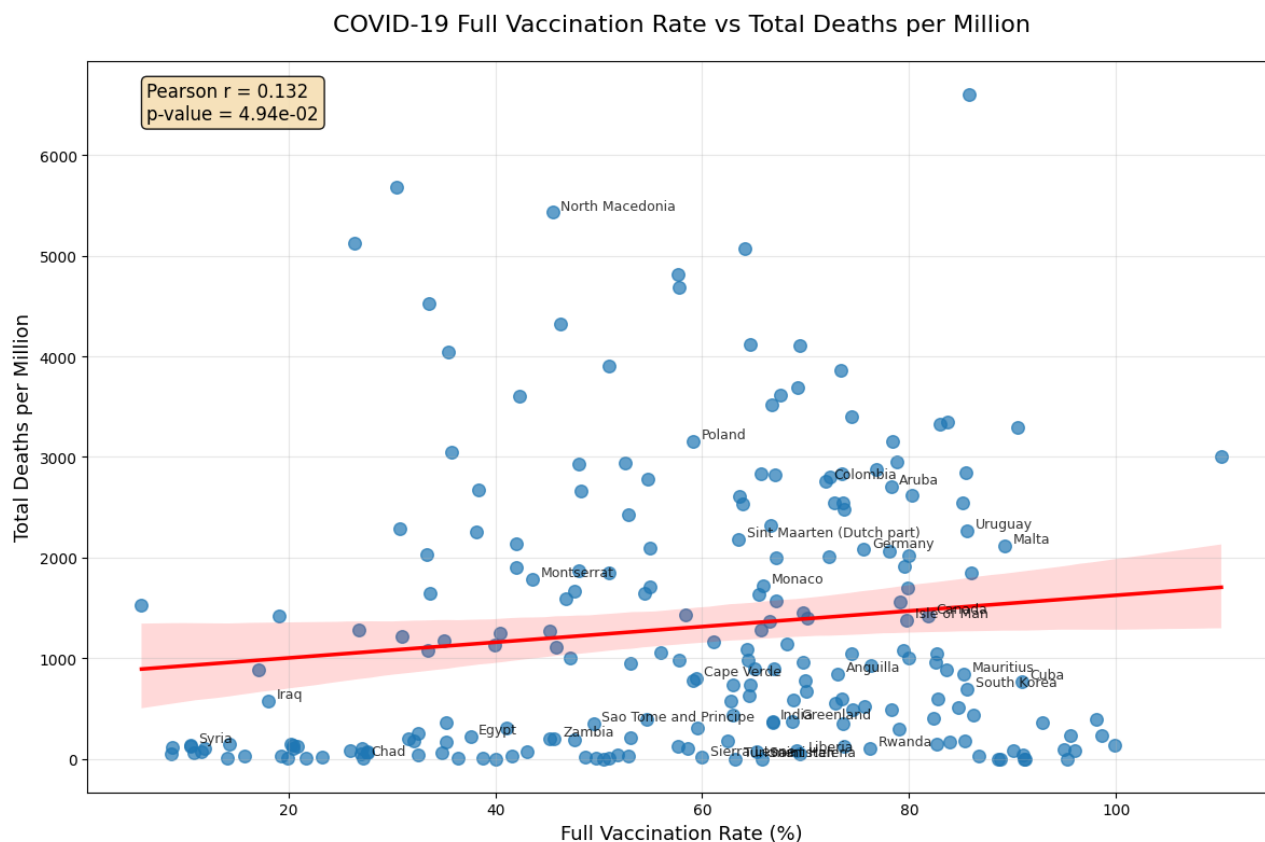
As shown in the scatter plot, countries with higher vaccination coverage tend to report slightly higher mortality (Pearson $r = 0.132$, $p = 0.049$).

This counter-intuitive result is largely explained by confounding factors: high-vaccination countries are typically more developed nations with advanced testing infrastructure, higher reporting accuracy, and older populations, leading to higher recorded death rates. In contrast, many low-vaccination countries show lower reported mortality, likely due to under-reporting.

No strong protective association observed at country level. The regression line shows only a modest upward trend, and the wide scatter indicates that vaccination rate alone explains little of the variation in mortality. Other factors such as demographics, healthcare quality, and pandemic timing play more significant roles.

Interpretation requires caution. These findings highlight the limitations of simple ecological correlations and the importance of contextual factors when evaluating vaccine impact across countries.





5.3. Correlation Analysis Between Economic Level and Vaccination Coverage

Economic level has a statistically significant positive impact on COVID-19 vaccination coverage

- The coefficient of determination R^2 is 0.28 for at least one dose coverage and 0.30 for full vaccination coverage, with both p-values < 0.001
- This indicates that per capita GDP can explain approximately 28%-30% of the global variation in vaccination coverage, and economic factors have a slightly greater impact on full vaccination than on the first dose
- The regression line slope shows that for every 10-fold increase in per capita GDP (1 unit increase in $\log_{10}$ GDP), first dose vaccination coverage increases by an average of about 20 percentage points

There is a significant "continental divide" in global COVID-19 vaccination, further corroborating the impact of economic factors

- The median first dose vaccination coverage by continent from highest to lowest is: South America (~83%) > Oceania (~80%) > Asia (~75%) > Europe (~72%) > North America (~68%) > Africa (~42%)
- Africa, as the continent with the lowest per capita GDP globally, also has a significantly lower median vaccination coverage than all other continents, which is highly consistent with the conclusions of the main correlation analysis
- Although South America's overall economic level is lower than that of Europe and North America, it achieved the highest median coverage through balanced vaccine promotion policies, providing strong evidence that economic factors are not the sole determinant

The spatial aggregation of economic levels directly leads to spatial inequality in vaccination coverage

- The three world maps present a highly consistent spatial pattern: high-income countries (North America, Western Europe, Australia, New Zealand, East Asia) almost all correspond to high vaccination coverage, while low-income countries (sub-Saharan Africa) almost all correspond to low vaccination coverage
- A clear "north-south difference" pattern has formed globally, with countries in the Northern Hemisphere generally having higher vaccination coverage than those in the Southern Hemisphere

Economic factors are not the only determinant of vaccination coverage

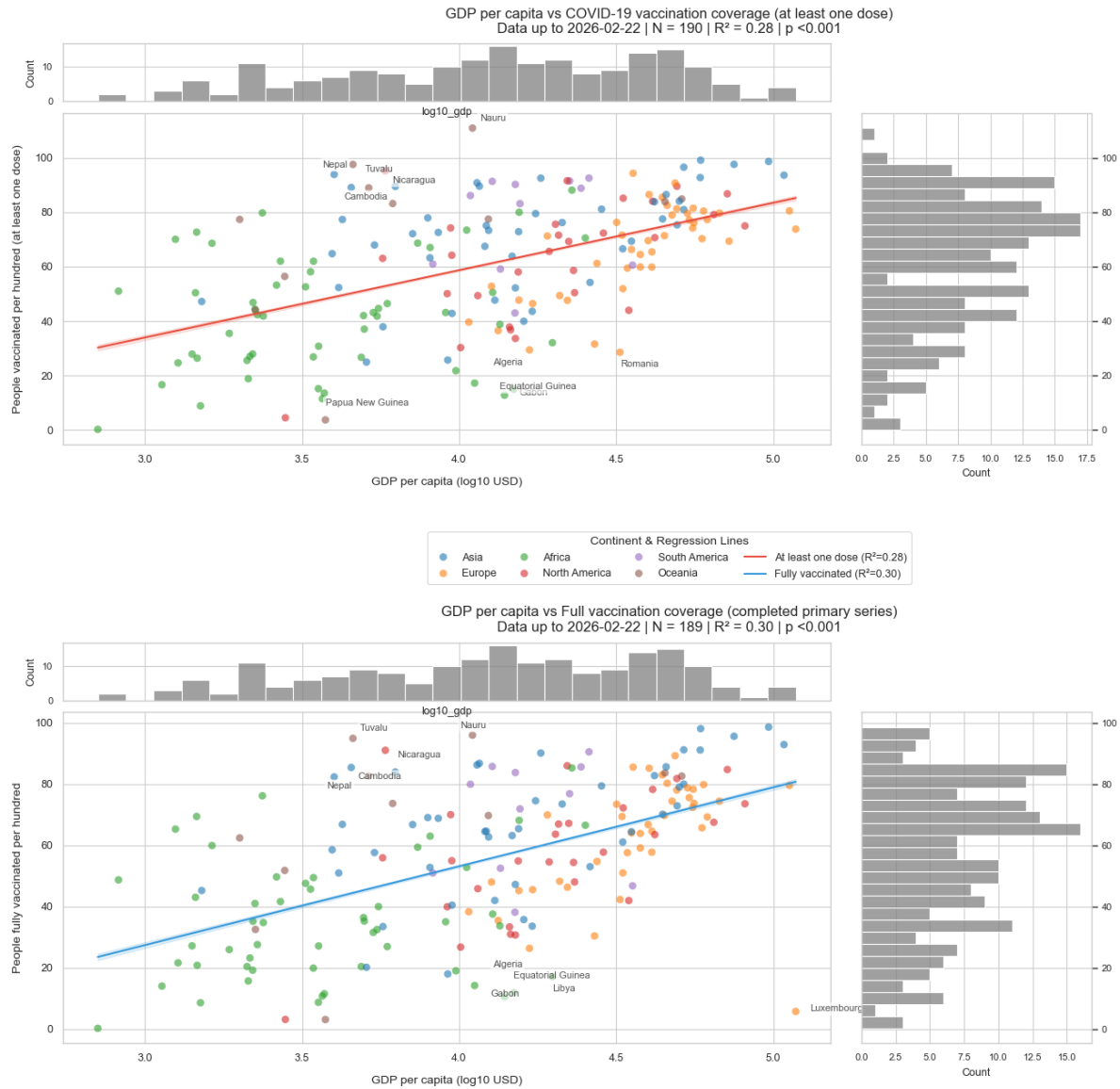
- Obvious outliers exist: middle-income countries such as China and Russia have vaccination rates among the highest in the world, reflecting the key role of strong government policy execution and public health systems
- Some high-income oil-producing countries in the Middle East have only moderate vaccination coverage, which may be affected by factors such as public vaccination willingness

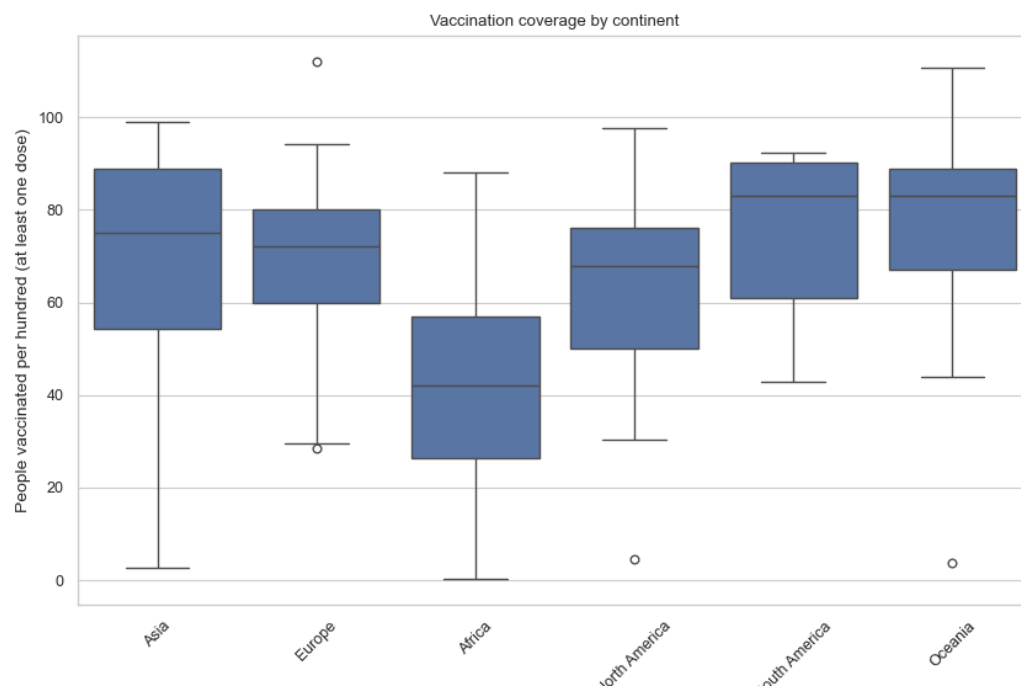
Low-income countries face a "double vaccination gap"

- Compared with high-income countries, low-income countries not only have lower first dose vaccination rates, but their full vaccination rates are further reduced by 10-20 percentage points relative to their first dose rates

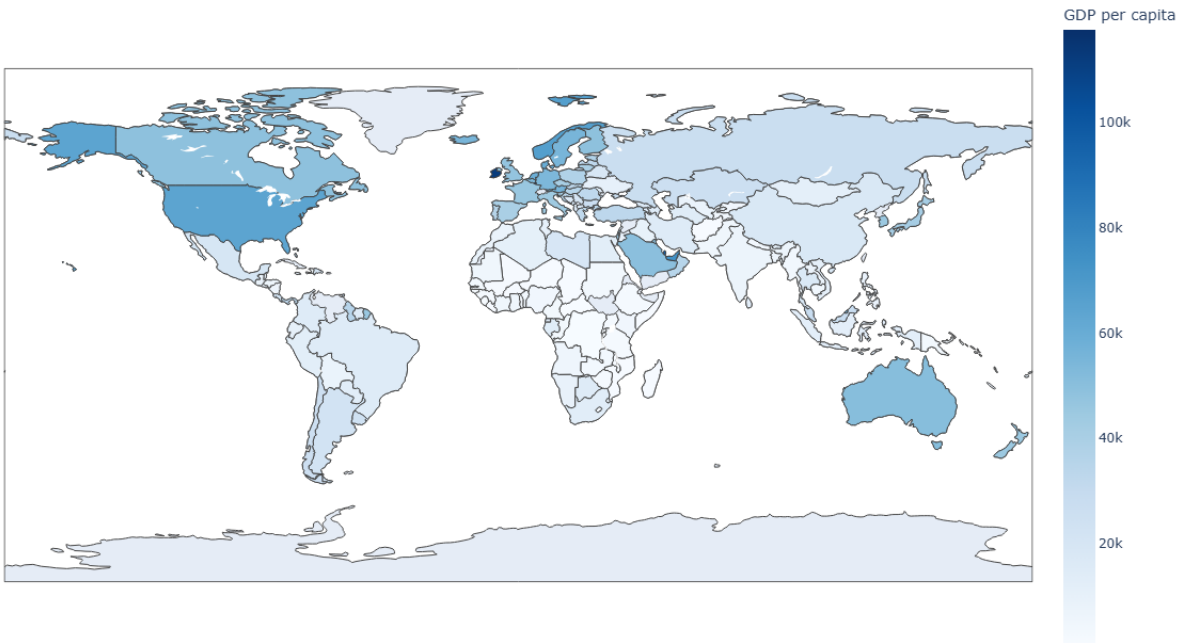


- This indicates that the "last mile" problem of vaccine promotion is more prominent in low-income countries, and they face greater challenges in completing the full vaccination process

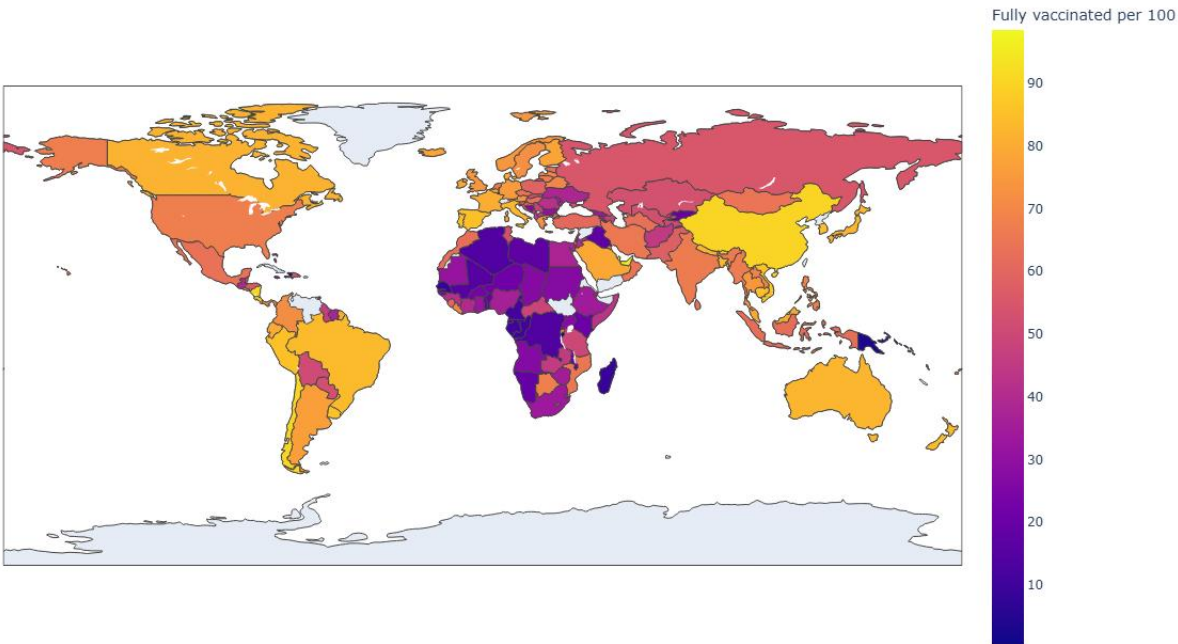




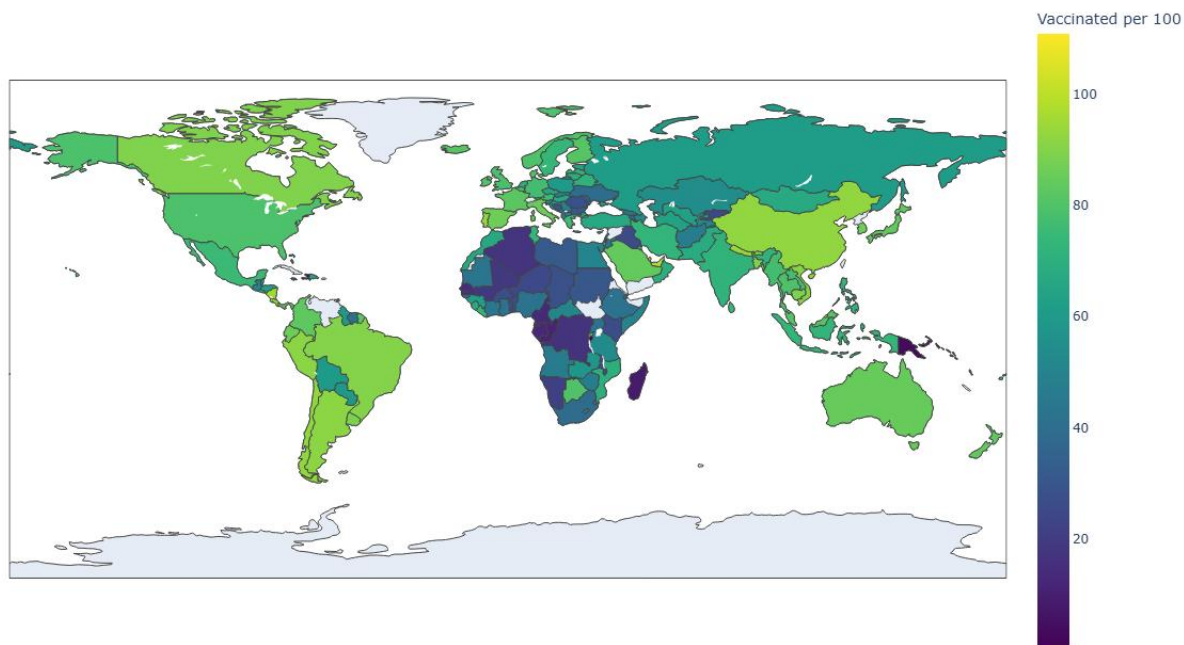
GDP per capita (USD)



Full Vaccination Coverage (completed primary series) per hundred people



COVID-19 Vaccination Coverage (at least one dose) per hundred people

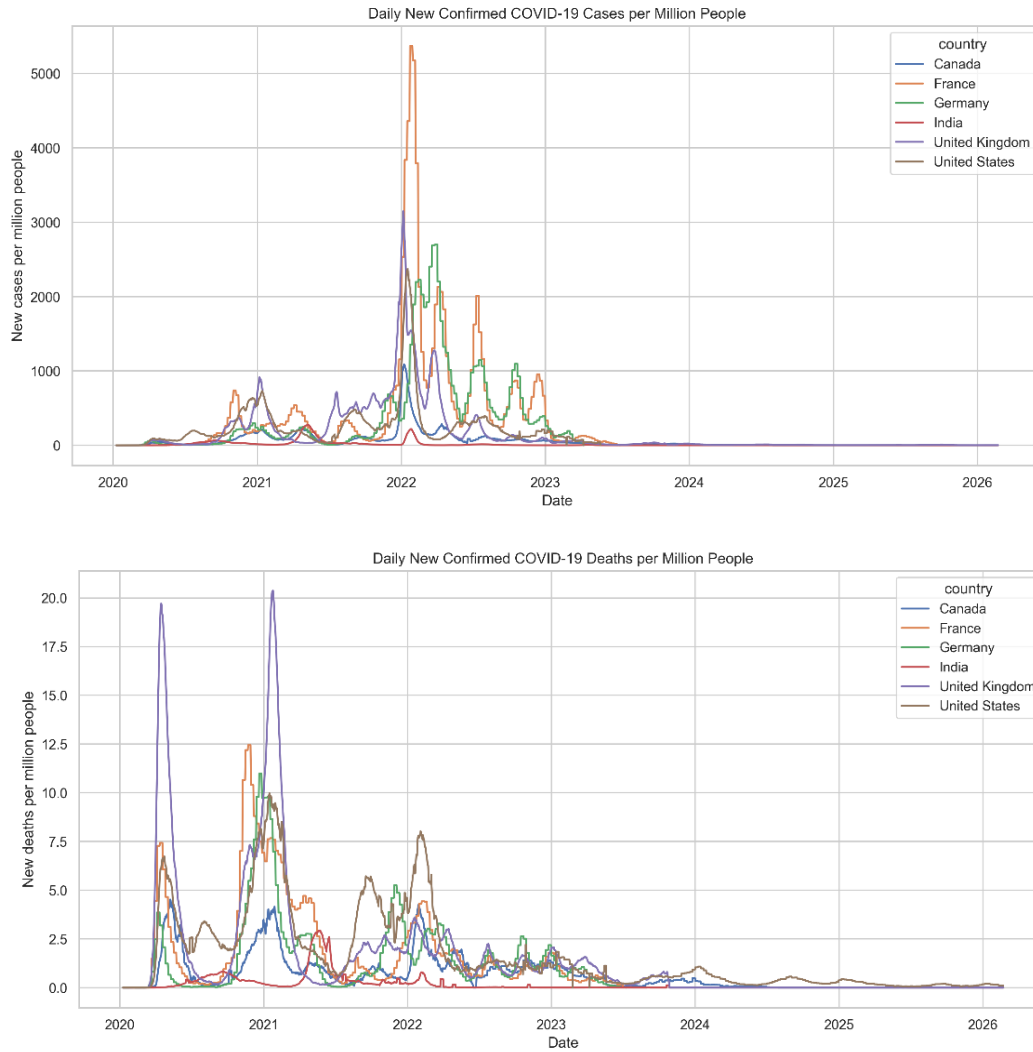


5.4. Global COVID-19 Time Trend Analysis

The time-series analysis revealed several major COVID-19 infection waves between 2020 and 2022 across the selected countries. The highest case peaks were observed around early 2022, especially in France, the United Kingdom, and Germany, which is consistent with the global spread of the Omicron variant. Compared with earlier waves, the Omicron period showed significantly higher confirmed case numbers per million people.

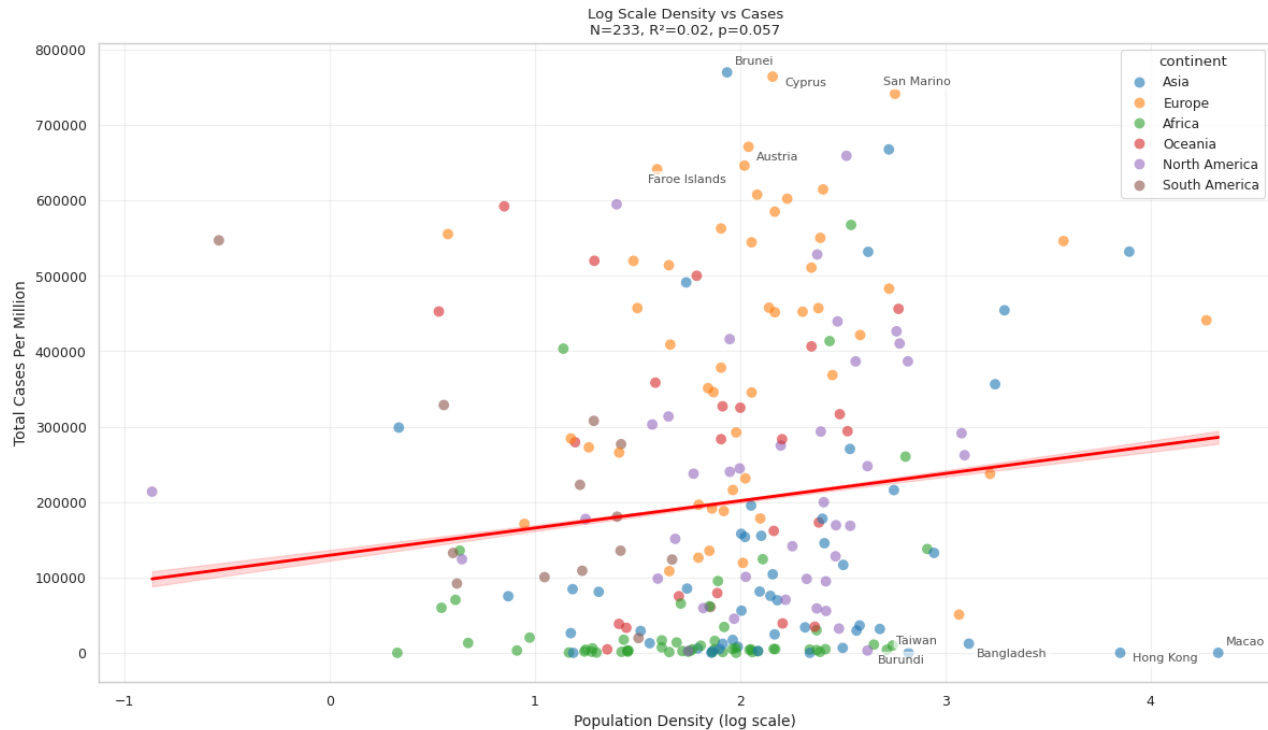
The death trend analysis showed that COVID-19 deaths generally lagged behind confirmed cases, indicating the delayed impact of infections on mortality. Although case numbers increased sharply during some periods, death rates after 2022 were relatively lower than earlier pandemic waves. This may be associated with vaccination rollout, improved healthcare responses, and changes in virus severity over time.

The analysis also demonstrated noticeable differences among countries. European countries such as France, Germany, and the United Kingdom experienced higher reported infection peaks, while countries such as India showed relatively lower reported values per million people. In addition, both confirmed cases and deaths gradually stabilized after 2023, suggesting a global transition from large-scale outbreak waves toward a more controlled stage of the pandemic.



5.5. Correlation Analysis Between Population Density and COVID-19 Cases

From the log-scale scatter plot and regression analysis of **population density vs. COVID-19 cases per million**, we derived the following key findings:



Weak Positive Correlation

There is a very weak positive relationship between population density and COVID-19 infection rates across countries. The regression line slopes slightly upward, indicating that countries with higher population density tend to report marginally higher confirmed cases per million people.

Low Explanatory Power

The coefficient of determination, $R^2 = 0.02$, shows that only 2% of the variation in COVID-19 cases can be explained by population density alone. This means density is not a strong predictor of outbreak severity.



Statistical Significance at the Marginal Level

The p-value of 0.057 is just above the common significance threshold of 0.05, meaning the observed trend is not statistically significant at the standard level. This reinforces that the relationship between density and cases is weak and inconsistent.

High Variability and Outliers

The plot shows a large spread of data points around the regression line, with many outliers. Countries like Brunei, Cyprus, and San Marino have high reported cases despite varying population densities, while some densely populated regions (e.g., Hong Kong, Macao, Bangladesh) show much lower case counts. This indicates other factors play major roles.

Regional Patterns

Countries in Europe and North America generally appear in the higher case count range, while many African countries show low infection rates regardless of density. This suggests regional differences in policy, healthcare access, and testing rates strongly influence reported outcomes.



6. Conclusion and Discussion

6.1. Core Conclusions

Based on comprehensive analysis of 190 countries' data up to February 22, 2026, this study draws four key conclusions:

1. **Economic strength is the strongest predictor of vaccination coverage**, explaining 28%-30% of global variation. Low-income countries face a "double gap" in both first-dose and full vaccination, creating a profound global vaccine equity divide.
2. **Non-economic factors are equally decisive**. South America achieved the highest median vaccination rate despite moderate economic levels, while some high-income countries underperformed, demonstrating that government policy and public health systems can offset economic disadvantages.
3. **Two common assumptions are not supported by country-level data**: Population density has only a negligible correlation with infection rates ($R^2=0.02$), and higher vaccination coverage does not correspond to lower reported mortality at the national level, due to confounding factors like testing capacity and age structure.
4. The pandemic exhibited clear **spatial-temporal clustering**, with distinct waves peaking during the Omicron variant in early 2022. Europe and the Americas were most severely affected in reported data, while Africa's low figures reflect both actual differences and significant underreporting.

6.2. Key Discussion

The counter-intuitive positive correlation between vaccination and mortality does not contradict individual-level vaccine efficacy. It instead highlights the limitations of ecological analysis: high-vaccination countries tend to have older populations and more accurate death reporting, while low-vaccination countries systematically undercount deaths. Similarly, the weak density-infection relationship shows that proactive public health measures can override inherent geographic risks, as demonstrated by successful control in many dense Asian cities.

The "continental divide" in vaccination coverage remains the most striking finding. Africa's median vaccination rate of 42%—less than half of South America's—exposes the failure of global solidarity during the pandemic. This gap not only prolonged the global crisis but also widened pre-existing health inequalities.



6.3. Limitations and Implications

This study is limited by cross-country data reporting biases, ecological fallacy risks, and the omission of variables like policy stringency and booster doses. Nevertheless, it provides clear policy lessons:

- Global vaccine equity must be a top priority for future pandemic preparedness
- Investing in robust public health systems yields greater returns than relying solely on economic strength
- Standardized global data reporting systems are essential for effective response

Future research should focus on longitudinal causal analysis of policy impacts and subnational-level studies to better understand within-country disparities. The COVID-19 pandemic has shown that global health security depends on protecting the most vulnerable, and the lessons learned must guide our response to future global health challenges.

